Reading the Radar

Concepts, math & answer key — instructor reference and participant handout

A hands-on guide to what weather radar sees — and misses · Clouds4Africa · July 2026

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How to use this reference. It mirrors the five notebooks. For every widget you get a short **concept**, the **math used in the code**, and worked **answers** at all three levels (Basic · A little further · Going deeper), followed by the radar-data case and references. Numbers match the widget defaults; where a value is read off a slider, the expected ballpark is given. Equations use plain notation so the page renders and prints anywhere.

THE FIVE BLOCKS

1. The beam in space
2. One measurement
3. Range, velocity & ambiguity
4. Wavelength, weather & the scan
5. Two polarizations

1 The beam in space

Where the echo comes from — a beam that spreads, climbs, bends, and can be blocked. *"No echo" is not "no weather."*

1 The beam spreads — beamwidth

The half-power beamwidth is set by the wavelength and the dish diameter. It is fixed in *degrees*, so the beam's physical cross-section grows linearly with range — two storms close together blur into one far away.

$$\lambda = c / f \quad \theta \approx 1.27\ \lambda/D \text{ (rad)} \quad \text{cross-beam width } w = r\cdot\theta \quad \text{gain } G \propto (D/\lambda)^2$$

NEXRAD: 2.8 GHz → λ=10.7 cm, D=8.5 m → θ≈0.92°; w≈1.6 km @100 km, 3.2 km @200 km. The 1.27 coefficient reflects a typical aperture taper (1.02 for uniform illumination).

BASIC

1. θ ≈ 0.92° (~0.9°). Width = r·θ → about 1.6 km at 100 km and 3.2 km at 200 km.
2. Wider. At 9.4 GHz with D=1.5 m, θ ≈ 1.5° — the much smaller dish dominates over the higher frequency, so the gap-filler's beam is broader than NEXRAD's.

A LITTLE FURTHER

1. The beam narrows. Beamwidth ∝ λ/D, so at fixed dish a shorter wavelength (C-band) sharpens angular resolution — less diffraction spreading for the same aperture.
2. Smaller. D = 1.27λ/θ with λ=5.35 cm, θ=0.9° → D ≈ 4.3 m. A shorter wavelength reaches the same beamwidth with a smaller dish.

GOING DEEPER

1. Doubling D (5→10 m) halves θ; gain ∝ $(D/\lambda)^2$ → 4× → **+6 dB**. A narrower beam concentrates power, buying both sensitivity and finer angular resolution.
2. width = r·θ = 2 km when r ≈ 2 km / 0.016 rad ≈ **125 km**. Cross-range resolution r·θ grows without bound because θ is fixed by the antenna, whereas along-range resolution Δr = cτ/2 is set by the pulse and stays constant — so a fixed dish can never hold cross-range resolution constant with range.

2 The beam climbs — height & refraction

Under standard refraction the beam bends slightly downward, but the Earth curves away faster, so the beam center *rises* with range. The beam also has a vertical depth ≈ $r \cdot \theta$. This is captured by the 4/3-Earth model.

$$R_e = (4/3) \cdot 6371 \approx 8495 \text{ km} \quad h = \sqrt{(r^2 + R_e^2 + 2r \cdot R_e \cdot \sin\varphi)} - R_e$$

$$\text{small-angle split: } h \approx r \cdot \sin\varphi + r^2/(2R_e)$$

0.5° tilt: h ≈ 1.9 km @120 km, 2.6 km @150 km, 4.1 km @200 km.

BASIC

1. ≈ 1.9 km at 120 km and ≈ 4.1 km at 200 km.
2. At 150 km the 0.5° center is ~2.6 km up — well above the surface, so shallow rain there is largely *under* the beam (overshooting).

A LITTLE FURTHER

1. At 1.5° the height at 150 km ≈ 5.3 km, about 2.6 km higher than 0.5°. Accept the higher beam when the low tilt is terrain-blocked — sampling high beats not sampling at all.
2. The ~2 km spread is the **beamwidth** (vertical extent $r \cdot \theta$ ≈ 2.4 km at 150 km), not refraction. Refraction and curvature set the center height; the band's thickness is the beam's angular width.

GOING DEEPER

1. Geometric rise $r \cdot \sin\theta$ ≈ 1.31 km; curvature $r^2/2R_e$ ≈ 1.32 km — about equal. They match when $r = 2R_e \cdot \sin\theta$ ≈ **148 km**. The tilt term (linear in r) dominates near the radar; the curvature term ($\propto r^2$) dominates far out.
2. The 0.5° center reaches 3 km at ≈ **165 km**; forced to 1.5° it reaches 3 km at ≈ **95 km** — blockage costs ~70 km of low-level QPE coverage.
3. Vertical depth ≈ $r \cdot \theta$ ≈ 2.4 km at 150 km (≈ 1.4–3.8 km). Any inversion or cold pool thinner than the beam depth is averaged with everything else in the beam — unresolvable, because the beam depth (not the electronics) sets vertical resolution.

3 The beam bends abnormally — anomalous propagation

The effective-Earth factor k is not constant — it depends on the vertical gradient of refractivity. Strong negative gradients (ducting) trap the beam near the surface and paint false echoes; weak gradients let it overshoot more.

$$R\_eff = a_e / (1 + a_e \cdot (dN/dh) \cdot 10^{-6}),\quad a_e = 6371\ \text{km},\quad k = R\_eff/a_e$$

standard $dN/dh \approx -39$ N/km → $k = 4/3$; ducting $dN/dh < -157$ N/km

modified refractivity $M = N + 0.157h$ → ducting ⇔ $dM/dh < 0$; strike range $r \approx -2 \cdot R\_eff \cdot \sin\theta$

BASIC

1. Standard dN/dh = −39 → k = 4/3 ≈ 1.33; the 150 km height matches Widget 2 (~2.6 km at 0.5°).
2. The effective radius collapses and the beam curves into the ground; the strike range follows $r \approx -2 \cdot R\_eff \cdot \sin\theta$ (tens of km, depending on tilt).

A LITTLE FURTHER

1. When R_eff → ∞, i.e. $1 + a_e \cdot (dN/dh) \cdot 10^{-6} = 0$ → $dN/dh = -10^6/a_e \approx$ **−157 N/km** — the ducting onset.
2. It takes strong super-refraction/ducting (more negative than −157). A nocturnal surface inversion does this; it forms after sunset and erodes after sunrise mixing, so AP clutter appears overnight and vanishes by mid-morning.

GOING DEEPER

1. dM/dh = dN/dh + 157 (N/km), so dM/dh < 0 ⇔ dN/dh < −157 — identical to R_eff → ∞. The slider's ducting onset matches.
2. $r \approx -2 \cdot R\_eff \cdot \sin\theta$ is linear in $\sin\theta$, so doubling elevation roughly doubles the strike range; it breaks down at large elevation/steep gradients where $\sin\theta \neq \theta$ and the curvature is no longer a small perturbation.
3. A ducting night does both: it bends part of the beam down into the ground at long range (AP clutter) and lowers the beam onto near terrain (new blockage). Both vanish once morning mixing destroys the inversion.

4 The beam gets blocked — terrain

Near-range terrain clips the lower beam. Per-gate blockage (PBB) is local; cumulative blockage (CBB) persists — every gate beyond a blockage inherits the loss.

$$\text{reflectivity bias} = 10{\cdot}\log_{10}(1 - \text{CBB})\ \text{dB} \quad \text{Z–R: } Z = 200{\cdot}R\text{^}1.6$$

PBB ≈ 50% when the terrain top reaches the beam-center height.

BASIC

1. Read the peak CBB at 0.5° off the widget (the default ridge gives a large fraction).
2. Raising the tilt clears the block once the lower beam edge rides above the ridge — typically a degree or two above 0.5°.

A LITTLE FURTHER

1. Power removed at the ridge is gone for every downrange gate — the beam never heals — so everything downrange of a blocked azimuth is biased low or absent.
2. A closer ridge at the same height blocks *more*: the beam is lower near the radar, so the ridge intercepts a larger fraction of it.

GOING DEEPER

1. bias = 10·$\log_{10}$(1−CBB). CBB = 0.7 → −5.2 dB → with R ∝ Z^(1/1.6), rain underestimated by ~54%. CBB = 0.5 → −3 dB → ~35% underestimate — large even for a 'half-blocked' gate.
2. PBB ≈ 50% when terrain top = beam-center height (half the beam below the obstacle). Read the 0.5° center height at 60 km from Widget 2, set the ridge there, confirm the PBB peak ≈ 50%.
3. Find the lowest tilt with CBB < 70%, then use Widget 2 to quantify how much higher you now sample at that azimuth — the trade is accepting higher-altitude sampling (overshooting low weather) to keep correctable data.

RD **Radar data — KNKX (San Diego), 11 March 2006**

Four tilts, reflectivity and velocity, same weather and radar position — only the tilt changes. Separate real weather from terrain/beam effects.

BASIC

1. The blocked wedge sits at a fixed azimuth (the terrain direction) and improves as you raise the tilt. A gap anchored to one direction is terrain blockage, not weather (which would move).

A LITTLE FURTHER

1. Raising the tilt lifts the beam over the ridge at that azimuth (Widgets 2–3), so the wedge fills in — proving it was blockage, not absent precipitation (real rain wouldn't reappear just by tilting up).

GOING DEEPER

1. The lowest unblocked tilt samples higher (estimate via Widget 2), missing shallow low-level weather there. A partial block biases all downrange reflectivity low (CBB), so every gate beyond the ridge inherits the underestimate.

References & going deeper

- Rinehart, *Radar for Meteorologists* — beam geometry, refraction, blockage (accessible).
- Doviak & Zrnić, *Doppler Radar and Weather Observations* — 4/3-Earth model, refractivity.
- Bech et al. (2003, *J. Atmos. Oceanic Technol.*) — beam-blockage simulation.
- Skolnik, *Radar Handbook* — antenna beamwidth and gain.

2 One measurement

How big a single value is, and whether it can be detected — a volume-sized average set by trading resolution, sensitivity, and range.

1 The pulse — one slider, three consequences

Pulse length τ sets range resolution, sensitivity, and blind range at once — coupled and pulling in opposite directions.

$\Delta r = c\tau/2$ (1.57 µs→235 m, 4.7 µs→705 m) sensitivity $\propto \tau$ (+3 dB per doubling) blind range $\approx c\tau/2$

BASIC

1. At 1.57 µs, $\Delta r \approx 235$ m < 400 m → resolved. They merge when $\Delta r \approx 400$ m, i.e. $\tau \approx 2{\cdot}400/c \approx 2.7$ µs; at merge $\Delta r \approx$ the 400 m spacing.
2. Blind range $\approx \Delta r$: ~235 m at 1.57 µs, ~705 m at 4.7 µs. The short pulse sees closer to the radar.

A LITTLE FURTHER

1. 1.57→4.7 µs is 3× τ → $+10{\cdot}\log_{10}(3) \approx +4.8$ dB. Choose the long pulse for weak clear-air echo; you give up range resolution (235→705 m) and a larger blind zone.
2. Targets just separate at spacing $\approx \Delta r = c\tau/2$; doubling τ doubles the merge separation, confirming the linear law.

GOING DEEPER

1. $\Delta r \approx 1$ km → $\tau \approx 6.7$ µs → $+10{\cdot}\log_{10}(6.7/1.57) \approx +6.3$ dB. Per doubling of τ: resolution coarsens ×2 while sensitivity gains +3 dB — opposite directions, so any single τ is a compromise. **Pulse compression** breaks the trade: a long coded pulse (long-pulse energy) compressed on receive to a short effective Δr.
2. At 4.7 µs the first ~700 m are blind, so a 500 m target is invisible. Clear-air mode accepts the large blind zone because it prioritizes sensitivity (long pulse) to detect very weak echoes; near-range blindness is an acceptable cost in fair weather.

2 The whole sample — the resolution volume

One measurement is a box: cross-section r·θ (grows with range) by depth cτ/2 (fixed). The volume grows as r^2, so distant pixels average huge volumes.

$$\text{width} = \text{height} = r\cdot\theta, \quad \text{depth} = c\tau/2, \quad \text{volume} \propto r^2$$

60→230 km: volume ratio = $(230/60)^2 \approx 14.7$.

BASIC

1. width = r·θ (0.92°): ≈ 0.96 km at 60 km, ≈ 3.7 km at 230 km (and equally tall).
2. Depth = cτ/2, set by the pulse-width slider; the range slider leaves it unchanged (~250 m everywhere).

A LITTLE FURTHER

1. The beam exceeds a 2 km core when r·θ > 2 km → r > ~125 km; beyond that the peak reads low (beam filling).
2. Volume ≈ $(r\cdot\theta)^2\cdot(c\tau/2)$: ~1.4 km^3 at 150 km vs ~3.4 km^3 at 230 km. The larger averaging volume smooths peaks, blends echo-top heights, and degrades QPE.

GOING DEEPER

1. Volume(230)/Volume(60) = $(230/60)^2 \approx 14.7$. A distant pixel averages ~15× more atmosphere than a near one — peaks smeared.
2. Vertical extent r·θ exceeds 500 m beyond r ≈ 31 km. So the bright band is a sharp ring near the radar (thin beam) but a broad smeared bias far out (beam thicker than the layer).

3 Can you even detect it? — the radar equation

Received power scales as $Pt \cdot D^2 \cdot \tau \cdot Z/(\lambda^4 r^2)$, so the minimum detectable Z climbs with range. Distributed weather falls only as r^2 (the volume fills with scatterers); a point target falls as r^4.

$$Pr \propto Pt \cdot D^2 \cdot \tau \cdot Z \,/\, (\lambda^4\, r^2) \;\; \Rightarrow \;\; \text{min-Z} \propto \lambda^4\, r^2 \,/\, (Pt \cdot D^2 \cdot \tau)$$

+6 dB per range doubling · 2× power = −3 dB · 2× dish = −6 dB · S→X at fixed dish ≈ −21 dB (λ^4)

BASIC

1. NEXRAD min-Z at 100 km ≈ −3 dBZ. Doubling transmit power improves it by 3 dBZ.
2. A +10 dBZ echo at 450 km is likely below the curve (undetected). The dish rescues it most (−6 dB per doubling, and it narrows the beam too); power is only −3 dB per doubling.

A LITTLE FURTHER

1. S→X at fixed dish improves min-Z by ~21 dB ($10 \cdot \log_{10}((\lambda_S/\lambda_X)^4) \approx 21$ dB) — matching $1/\lambda^4$.
2. A 2 m X-band dish vs the 8.5 m NEXRAD at 100 km: dish penalty $10 \cdot \log_{10}((2/8.5)^2) \approx -12.6$ dB, wavelength gain ≈ +21 dB → net ~+8 dB *more* sensitive (min-Z ≈ −11 vs −3 dBZ). A quarter-size dish beats NEXRAD because the short wavelength more than compensates. The price, in Block 4: attenuation.

GOING DEEPER

1. Distributed weather fills the resolution volume, which grows as r^2, partly offsetting the $1/r^4$ power loss → net $1/r^2$ (min-Z $\propto r^2$, +6 dB/doubling). A point target doesn't grow with the volume → full $1/r^4$ (+12 dB/doubling). The r^2 falloff is gentler, so weak distant rain is comparatively easier to detect than a weak point target at the same range.
2. Pick τ that just detects 0 dBZ at 200 km; the price is coarser Δr (Widget 1) and a deeper sample (Widget 2). Good for clear-air mode (sensitivity matters, resolution doesn't); poor for severe convection (you need fine resolution for cores and shear).

RD Radar data — Moore, 20 May 2013: one storm, two ranges

The same storm at the same time from KTLX (~20 km) and KINX (~200 km), both centered on the storm — the identical ground sampled by a small volume and a large one.

BASIC

1. The near radar (KTLX) is sharper — tighter core, finer structure, higher peak. The far radar (KINX) is bigger, blurrier, with a lower peak.
2. The resolution volume ($r \cdot \theta$): KINX's beam is ~10× wider, averaging the storm over far more area.

A LITTLE FURTHER

1. Beam width ≈ 0.3 km (KTLX) vs ~3.3 km (KINX). Sampling higher can't make an echo physically wider, so the broadening proves beam-width smearing (resolution / beam filling), independent of height.
2. KINX shows less of the weak periphery: its higher minimum detectable Z ($\propto r^2$) drops the faint edges below the noise floor, while KTLX still catches them.

GOING DEEPER

1. KINX's 0.5° beam at 200 km is ~4 km up; the KTLX tilt also reaching ~4 km at 20 km is ~12°. With both at the same height, the residual sharpness gap is pure beam width — the far storm still looks broader.
2. Volume ratio $\propto$ (range ratio)^2 ≈ 10^2 = ~100×. One KINX pixel averages ~100× more air; smearing the storm across that much atmosphere lowers the peak reflectivity.

References & going deeper

- Rinehart, *Radar for Meteorologists* — pulse volume, radar equation, sensitivity.
- Doviak & Zrnić — resolution volume, weather radar equation, beam filling.
- Fabry, *Radar Meteorology: Principles and Practice* — intuitive treatment of sampling.
- NWS/WDTD training (training.weather.gov) — beam filling and range effects.

3 Range, velocity & ambiguity

How far and how fast — and where sampling breaks. One pulse train, one wall ($c\lambda/8$); folding and aliasing are artifacts you learn to read.

1 Range folding

A pulse must return before the next goes out; beyond r_max it is misplaced to a shorter range as a second-trip echo.

$$r\_max = c / (2 \cdot PRF)$$ (1000 Hz → 150 km) · apparent range = true − n·r_max

BASIC

1. 1000 Hz → r_max = 150 km. A target at 320 km wraps to 320 − 2·150 = **20 km**.
2. Lowering the PRF grows r_max (inverse relation).

A LITTLE FURTHER

1. r_max ≥ 300 km needs PRF ≤ c/(2·300 km) = 500 Hz; that costs Nyquist v = λ·PRF/4 ≈ 13 m/s (low — heavy aliasing).
2. A second-trip echo arrives one inter-pulse period late, so the radar assigns a travel time short by exactly the pulse spacing = r_max in range — hence true − r_max.

GOING DEEPER

1. Surveillance (low PRF, long r_max, no velocity) plus Doppler (high PRF, short r_max, velocity). Two sweeps per tilt cost scan time, slowing the volume — tied to Block 4's update-time trade.
2. 380 km at 1000 Hz → 380 − 2·150 = 80 km displayed. Distinguish from a real 80 km storm by continuity between scans and by changing the PRF: true storms move coherently and stay put; second-trip echoes relocate or vanish when the PRF changes.

2 The Doppler dilemma

The PRF sets r_max and v_max in opposite directions; their product is fixed by wavelength alone — a hard wall a single PRF can only slide along.

$$r\_max \cdot v\_max = c \cdot \lambda / 8 \;\cdot\; \text{S-band} \approx 4013\ \text{km}\cdot(\text{m/s}) \;\cdot\; v\_nyq = \lambda \cdot PRF/4$$

BASIC

1. No — a single PRF only moves the operating point *along* the curve; it never leaves it.
2. r_max·v_max stays constant (~4013 km·m/s at S-band) as you slide.

A LITTLE FURTHER

1. No: 35 m/s × 150 km = 5250 > 4013, so it's unreachable at any single S-band PRF — short by ~1.3× (≈30%).
2. Harder. C-band's shorter wavelength halves cλ/8, so the product ceiling halves (≈2× worse).

GOING DEEPER

1. The product ratio S:C = λ_S/λ_C ≈ 2, so C-band makes the dilemma ~2× worse — a real constraint for Africa's C-band networks.
2. 3:2 dual-PRF (1200/800 Hz): $v_1 \approx 32$ m/s @125 km, $v_2 \approx 21$ m/s @187 km; $v\_ext = v_1 v_2/(v_1 - v_2) \approx$ **64 m/s**. A single PRF reaching 64 m/s would need ~2400 Hz → r_max ≈ 63 km. Dual-PRF delivers 64 m/s while keeping ~125–187 km — *beyond* the single-PRF curve — bringing the 35 m/s @150 km storm within reach.

3 Velocity aliasing

Velocity beyond ±v_nyq folds into the window — a fast outbound storm can read inbound, producing sharp unphysical jumps on the display.

$$v\_nyq = \lambda\cdot PRF/4 \;\cdot\; \text{fold: } v\_disp = ((v+v\_nyq) \bmod 2v\_nyq) - v\_nyq \;\cdot\; \text{de-alias: add } 2\cdot v\_nyq$$

BASIC

1. 1000 Hz, S-band → v_nyq ≈ 26.8 m/s. A 35 m/s storm folds to 35 − 2·26.8 ≈ **−18.5 m/s** (reads inbound).
2. Need v_nyq ≥ 35 → PRF ≥ 4·35/0.107 ≈ 1308 Hz → r_max ≈ 115 km — you gave up range (150→115 km).

A LITTLE FURTHER

1. Aliasing. A +25 to −25 jump across a smooth gradient implies 50 m/s of real shear in one gate — unphysical; the widget shows the fold produces exactly this ±2v_nyq jump.
2. 3:2 dual-PRF lifts v_nyq to ~64 m/s, so 35 m/s no longer folds, and the ~187 km range covers 150 km — capturing the storm no single PRF could.

GOING DEEPER

1. True +40 at v_nyq = 28: displayed = ((40+28) mod 56) − 28 = 12 − 28 = **−16 m/s**. De-alias: −16 + 2·28 = +40 m/s. An algorithm makes the same call by continuity with neighbouring gates.
2. Reflectivity (power) never folds; only range and velocity are ambiguous, trading via cλ/8. The low-PRF surveillance sweep gives long range but a Nyquist too low for usable velocity; the high-PRF Doppler sweep gives good velocity only at short range — the same dilemma from the data side.

RD Radar data — Hurricane Isabel, KAKQ, 18 September 2003

Reflectivity (left) and base velocity (right) at the same scale. Read the patterns, not individual gates.

BASIC

1. The red/green split is the real circulation — wind blowing away (red) on one side, toward (green) on the other, with a near-zero curve through the centre.
2. Velocity covers a smaller disk than reflectivity, because the high-PRF Doppler sweep has a shorter r_{max} than the low-PRF surveillance sweep; the rainbands reach much farther in reflectivity.

A LITTLE FURTHER

1. An abrupt opposite-extreme patch dropped into a smooth region is aliasing, not a real reversal — no storm feature justifies a sudden full flip.
2. Reflectivity (low PRF, long range) places the distant rain; velocity (high PRF, short range) can't reach it. A *lower* PRF would extend the velocity range but lower the Nyquist (more aliasing) — the dilemma.

   (A higher PRF would shrink the range, not extend it.)

GOING DEEPER

1. The velocity-disk edge is the Doppler r_{max}. Because $r_{max} \cdot v_{max} = c\lambda/8$, the high-PRF velocity scan necessarily has short range, while low-PRF reflectivity reaches far but carries no velocity.
2. 3:2 dual-PRF (1200/800 Hz) lifts the Nyquist to ~64 m/s and keeps the lower PRF's ~187 km range, so a 35 m/s @150 km storm neither folds nor runs out of range — captured where a single PRF couldn't.

References & going deeper

- Doviak & Zrnić — range/velocity ambiguity, the $c\lambda/8$ relation, dealiasing.
- Rinehart, *Radar for Meteorologists* — PRF, Nyquist, second-trip echoes.
- Torres & Curtis — staggered-PRT and dual-PRF techniques (NSSL/NWS literature).
- NWS/WDTD velocity training — reading aliasing and range folding operationally.

4 Wavelength, weather & the scan

What the wavelength decides, and how a volume is assembled — sensitivity vs attenuation, and detail vs velocity vs update speed.

1 Shorter wavelength, bigger price — attenuation

Rain absorbs and scatters energy out of the beam, so everything behind a cell is seen through a curtain. The loss is negligible at S, real at C, severe at X — and it is two-way and path-integrated.

specific attenuation $k = a·R^b$ dB/km (a: S≈1e-4, C≈6e-3, X≈3e-2; b≈1) · two-way PIA = 2∫k dr · $Z = 200·R^{1.6}$

BASIC

1. S-band reads the peak closest to truth; X-band loses everything behind the core.
2. At 80 mm/h the C-band echo loses roughly 16 dBZ behind the cell (read off the widget).

A LITTLE FURTHER

1. A C-band shadow underestimates rainfall downrange of a heavy core — a town behind the core has its rain undercounted.
2. The X-band shadow *accelerates* with rain rate (a path integral of k ∝ R), so heavy convection compounds the loss rapidly.

GOING DEEPER

1. PIA = 2∫k dr grows super-linearly as the cell intensifies, so a second cell *behind* the first is essentially invisible at X-band — its echo is extinguished in transit.
2. S-band needs no correction (NEXRAD); C-band does. A ~16 dB two-way C-band deficit, via R ∝ Z^(1/1.6), is roughly a 90% rainfall underestimate downrange (illustrative).

2 Sensitivity vs attenuation — which band sees the distant storm?

Two wavelength effects oppose: shorter λ is intrinsically more sensitive ($1/\lambda^4$, a fixed head-start) but attenuates more (a penalty that grows with range). They cross at a crossover range.

intrinsic gap X vs S ≈ +21 dB ($1/\lambda^4$, range-independent) · attenuation penalty = $2\int k\,dr$ (grows with range)

BASIC

1. Rain = 0: X-band detects the weakest echoes at every range ($1/\lambda^4$ head-start, no attenuation).
2. Raising the rain, the X-band curve climbs fastest (most attenuation); S-band barely moves.

A LITTLE FURTHER

1. X beats S inside the crossover range, S beyond it. In one sentence: the shorter wavelength wins near the radar (sensitivity) but loses far away (accumulated attenuation).
2. Heavier rain moves the crossover toward the radar (the penalty accumulates faster), shrinking X-band's useful range.

GOING DEEPER

1. A fixed +21 dB head-start is constant with range, but the attenuation penalty is a path integral that grows without bound — so beyond some range it always overtakes the head-start, and the longer wavelength wins for distant weak rain.
2. A compact C-band radar is more sensitive than NEXRAD at the same dish yet sees less through heavy rain. Band choice trades sensitivity against penetration; polarimetric attenuation correction (KDP, Block 5) restores lost signal and pushes the effective crossover outward.

3 Putting it together — scan strategies

A VCP is a stack of tilts, trading the cone of silence (above the top tilt), the low-level gap (below the bottom tilt), and the update time ($\propto$ number of tilts).

volume time = N_tilts·(60/rpm) · VCP21 (9 tilts, 3 rpm) = 180 s · VCP12 (14 tilts) = 280 s · cone half-angle = 90° − (top tilt)

BASIC

1. VCP21 (9 tilts) = 180 s vs VCP12 (14 tilts) = 280 s at 3 rpm — VCP21 is faster (fewer tilts).
2. At 30 km, a top tilt of ~19.5° reaches ~10 km; a storm top above that sits in the cone of silence and is missed.

A LITTLE FURTHER

1. Under 3 min at 3 rpm allows ≤ 9 tilts (9·20 s = 180 s). More tilts exceeds 3 min; the cost is vertical coverage (a bigger cone and gap).
2. 25 s/tilt × 14 = 350 s (vs 280). This velocity-vs-time trade motivates electronically-steered **phased-array** radar.

GOING DEEPER

1. Build a VCP for a fast severe line: weight low tilts (low-level detection), keep enough mid tilts for structure, accept a larger cone, and hold the volume time short for rapid updates — each trade justified against detection, velocity quality, and coverage.
2. Cone half-angle = 90° − top tilt (≈ 70.5° for a 19.5° top). That top tilt reaches 12 km at r ≈ 33 km, so between the radar and ~33 km you're blind to 12 km tops — an argument for a higher top tilt or a faster-revisiting phased array.

RD Radar data — S/C pairs: the wavelength trade, both ways

Two pairs, each the same storm on an S- and a C-band radar: a heavy core (where C loses to attenuation) and a lighter echo (where C wins on sensitivity).

BASIC

1. Behind the strongest core, C-band shows weaker echo than S (treated as truth) — that deficit is attenuation; C loses.
2. In the light-echo pair, OU-PRIME (C) fills the weak field while KOUN (S) breaks it into speckle — C wins on sensitivity at a similar dish.

A LITTLE FURTHER

1. Shorter λ is more sensitive (light-echo) but attenuates more (heavy-core): C beats S for faint near echo and loses behind heavy cores — the detectability crossover.
2. Estimate the C deficit by comparing dBZ categories (often ~15–20 dB) — consistent with a heavy cell; via $Z = 200 \cdot R^{1.6}$ that is a large rainfall underestimate for a downrange town.

GOING DEEPER

1. One C-band radar out-detects S in light rain (sensitivity, near) yet goes blind behind a heavy core (attenuation, far). A bigger/quieter receiver lowers the detection threshold (helps faint distant rain) but cannot recover signal already absorbed in transit — a detection-threshold problem versus a signal-removed-in-transit problem.
2. The light-echo pair confounds wavelength with dish, power, and receiver differences, so the sensitivity 'win' isn't cleanly attributable to wavelength. The heavy-core attenuation conclusion is robust because attenuation is a large, calibrated, wavelength-dominated path loss that extra power or gain can't undo.

   If you add the radar specs (C: 0.5°, 1 MW; S: 1°, 500 kW), the sensitivity pair becomes *fully* confounded — power +3 dB, finer beam +6 dB, and λ^2 +6 dB all favour C — while the attenuation conclusion still holds.

References & going deeper

- Rinehart, *Radar for Meteorologists* — wavelength bands, attenuation, scan strategies.
- Doviak & Zrnić — attenuation, the weather radar equation, VCP design.
- Carey et al. (2000, *J. Appl. Meteor.*) — C-band attenuation correction via differential phase.
- NWS VCP documentation (training.weather.gov) — VCP 12/21/35, cone of silence.

5 Two polarizations

What the echo is made of — ZDR (shape), ρhv (variety), KDP (attenuation-proof rain). Take-home bonus block.

1 Shape — differential reflectivity (ZDR)

Bigger raindrops flatten as they fall, so the horizontal wave sees a wider target and the H echo is stronger. ZDR reads shape and size — independent of how many drops there are.

axis ratio $\gamma \approx 1.03 - 0.062 \cdot D$ (D in mm) · $\mathrm{ZDR} = 20 \cdot \log_{10}((L\_z+k)/(L\_x+k))$, k = 1/(ε−1) ≈ 1/79

L_z, L_x = oblate-spheroid depolarization factors. ZDR(D): 1 mm→0.3, 3 mm→1.7, 5 mm→3.2, 6 mm→4.1 dB. Z ∝ $N \cdot D^6$.

BASIC

1. 1→6 mm: the drop flattens (γ ≈ 0.97→0.66) and ZDR rises from ~0.3 to ~4.1 dB.
2. Changing log N changes Z (∝ N) but leaves ZDR unchanged — ZDR measures shape and size, not amount.

A LITTLE FURTHER

1. A small-drop drizzle (ZDR ≈ 0) and a big-drop shower (high ZDR) can share the same Z; ZDR separates them. It matters because rain rate depends on the drop-size distribution — same Z, different R.
2. Tumbling hail looks round on average → ZDR ≈ 0, even at high Z — unlike a big raindrop (high ZDR). That contrast flags hail.

GOING DEEPER

1. +1 in log N raises Z by 10 dB with ZDR unchanged. A ratio (intensive) cancels absolute calibration and concentration, so it's robust to miscalibration and beam filling; Z (extensive) scales with both and needs calibration.
2. Z depends on size and number; ZDR pins the size. Combine them: ZDR gives the drop size, then Z fixes the concentration — two equations, two unknowns, which single-pol Z alone cannot solve.

2 Variety — the correlation coefficient (ρhv)

ρhv measures how alike the H and V returns are across the sample volume: ≈1 for a uniform population (pure rain), lower for mixtures, very low for non-weather. A purity meter.

$$\rho_{hv} = |\langle s\_h \cdot s\_v^* \rangle| \,/\, \sqrt{(\langle |s\_h|^2 \rangle \langle |s\_v|^2 \rangle)}$$ · regimes: >0.97 rain · 0.85–0.97 mixed/hail/melt · <0.85 non-met

BASIC

1. Low variety → ρhv ≈ 1.0 → a uniform target, i.e. pure rain.
2. Raising variety spreads the cloud and lowers ρhv; the label switches from rain to a mixture around ρhv ≈ 0.97.

A LITTLE FURTHER

1. A melting layer (bright band) and rain+hail mixtures both give intermediate ρhv (~0.95).
2. Birds, insects, ground clutter, and chaff are irregular and randomly oriented → very low ρhv (below ~0.8), which flags them where Z and velocity can't.

GOING DEEPER

1. A melting layer mixes water-coated snowflakes of many shapes and sizes → high variance in the H/V ratio → low ρhv. It is cleaner than Z because the bright-band Z enhancement is modest and ambiguous, while ρhv drops sharply and specifically.
2. A huge distant resolution volume (Block 2) sweeps more particle types into one gate, lowering ρhv by beam broadening alone — not because the weather changed — so ρhv must always be read alongside the sample size.

3 Phase — specific differential phase (KDP) and rainfall

The H wave is slowed more than the V wave through flattened rain, so a differential phase ΦDP accumulates; its slope KDP is proportional to rain rate. Being a phase, KDP is immune to attenuation and calibration — the answer to Block 4's shadow.

$$KDP = (R/44)^{(1/0.82)}\ \text{deg/km} \cdot R = 44 \cdot KDP^{0.82}\ \text{(exact inverse)} \cdot \Phi DP = 2 \cdot \int KDP\ dr$$

KDP @50 mm/h ≈ 1.17 deg/km. Z is attenuated behind a cell; KDP is not.

BASIC

1. Raising the peak rain: observed Z behind the cell drops (attenuation, Block 4), while ΦDP keeps climbing (it accumulates, unaffected).
2. R-from-KDP tracks the truth; R-from-Z reads biased low behind the cell.

A LITTLE FURTHER

1. At X-band the Z-based rain rate worsens (more attenuation), but KDP-based stays correct because phase isn't attenuated — only the slope of ΦDP matters, and ΦDP still accumulates.
2. ΦDP only accumulates (monotonic), so it's immune to the multiplicative power errors — attenuation, miscalibration, partial beam blocking — that knock Z down; those don't change the differential phase.

GOING DEEPER

1. KDP needs ΦDP to change across several gates, so it's noisy in light rain and undefined in none, but excellent in heavy rain (large ΦDP slope) — exactly where Z-based QPE fails most. Blend: Z/ZDR for light rain, KDP for heavy rain (composite R(Z, ZDR, KDP) estimators).
2. Behind a heavy C-band core, Z is attenuated and ZDR is reduced by differential attenuation — both power measures fail — yet KDP (phase) is untouched, which makes it the workhorse for rainfall on C-band networks.

RD Radar data — KTLX, Moore, 20 May 2013 (Z, ZDR, ρhv, ΦDP)

Four panels of one storm. Read the patterns, and expect real data to be messier than the widgets.

BASIC

1. A hail core reads high Z, low ZDR (tumbling stones look round), and reduced ρhv (a chaotic mix). Here it's ambiguous — the cores are far (~100 km) and high on the 0.48° beam.
2. Moderate Z with high ZDR is big, flattened drops; Z alone can't separate that from a hail core (similar Z), but ZDR (high vs ~0) can.

A LITTLE FURTHER

1. A ρhv ring at fixed range marks the melting layer; ρhv flags it cleanly (a sharp drop) because mixed-phase particles decorrelate H and V, whereas Z only brightens modestly.
2. Very low ρhv with low Z away from the storm is birds/insects/clutter; single-pol can't easily flag it (it looks like weak Z), but ρhv exposes it.

GOING DEEPER

1. Classify by Z–ZDR–ρhv: heavy rain (high Z, moderate ZDR, ρhv≈1), big drops (moderate Z, high ZDR), hail (high Z, low ZDR, lowered ρhv), melting layer (ρhv ring), non-met (low ρhv) — the logic a hydrometeor-classification algorithm automates.
2. Behind an attenuated core, use KDP (immune) for the rain rate while ZDR and ρhv classify the echo. Dual-pol is decisive for C-band because it both identifies and corrects what attenuation corrupts.

References & going deeper

- Bringi & Chandrasekar, *Polarimetric Doppler Weather Radar* — ZDR, ρhv, KDP theory.
- Ryzhkov & Zrnić, *Radar Polarimetry for Weather Observations* — modern, applied.
- Kumjian (2013, *J. Operational Meteor.*) — dual-pol variable interpretation (accessible).
- NWS/WDTD dual-pol training (training.weather.gov) — operational signatures.
- Park et al. (2009) / Ryzhkov et al. — hydrometeor classification and attenuation correction.
- Py-ART documentation (arm-doe.github.io/pyart) — computing and displaying KDP.